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Calibration results
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Normalized Residuals
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Reprojection error (cam0):     mean 0.17206420107807036, median 0.1398585400019785, std: 0.12460015187357101
Reprojection error (cam1):     mean 0.1736932272302403, median 0.13931786003653485, std: 0.12832228071314744
Gyroscope error (imu0):        mean 0.24867161207206287, median 0.23228235983247272, std: 0.12567779381398334
Accelerometer error (imu0):    mean 0.1776438655969784, median 0.15098875506653112, std: 0.12540630673011974

Residuals
----------------------------
Reprojection error (cam0) [px]:     mean 0.17206420107807036, median 0.1398585400019785, std:
0.12460015187357101
Reprojection error (cam1) [px]:     mean 0.1736932272302403, median 0.13931786003653485, std:
0.12832228071314747
Gyroscope error (imu0) [rad/s]:     mean 0.0013355541563899182, median 0.0012475315076994667, std:
0.0006749845649673083
Accelerometer error (imu0) [m/s^2]: mean 0.0213371082103796, median 0.018135517342963488, std:
0.015062765764370614

Transformation (cam0):
-----------------------
T_ci:  (imu0 to cam0):
[[ 0.02961627 -0.99955989  0.00170214  0.01594428]
[-0.02012616 -0.00229886 -0.99979481  0.0315593 ]
[ 0.9993587   0.02957594 -0.02018538 -0.09218218]
[ 0.          0.          0.          1.        ]]

T_ic:  (cam0 to imu0):
[[ 0.02961627 -0.02012616  0.9993587   0.09228602]
[-0.99955989 -0.00229886  0.02957594  0.01873619]
[ 0.00170214 -0.99979481 -0.02018538  0.02966495]
[ 0.          0.          0.          1.        ]]

timeshift cam0 to imu0: [s] (t_imu = t_cam + shift)
-0.01228894370220883


Transformation (cam1):
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```
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T_ci:  (imu0 to cam1):
[[ 0.02910796 -0.99957532  0.00138207 -0.0340443 ]
[-0.02230088 -0.00203172 -0.99974924  0.03174362]
[ 0.99932747  0.02906984 -0.02235055 -0.09187432]
[ 0.          0.          0.          1.        ]]

T_ic:  (cam1 to imu0):
[[ 0.02910796 -0.02230088  0.99932747  0.09351141]
[-0.99957532 -0.00203172  0.02906984 -0.03129458]
[ 0.00138207 -0.99974924 -0.02235055  0.02972928]
[ 0.          0.          0.          1.        ]]

timeshift cam1 to imu0: [s] (t_imu = t_cam + shift)
-0.012266394287608385

Baselines:
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Baseline (cam0 to cam1):
[[ 0.99999982  0.00033026 -0.00050198 -0.05004528]
[-0.00033135  0.9999976  -0.00216635 -0.00001002]
[ 0.00050126  0.00216651  0.99999753  0.00023126]
[ 0.          0.          0.          1.        ]]
baseline norm:  0.050045814036265006 [m]


Gravity vector in target coords: [m/s^2]
[-0.19779856 -9.8045482  -0.01154408]


Calibration configuration
=========================

cam0
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```

Camera model: pinhole
Focal length: [387.26496788634023, 388.0178836701606]
Principal point: [320.53538193017073, 238.7961117373796]
Distortion model: radtan
Distortion coefficients: [-0.0048964025986568405, -0.0032256738628326904, 6.867803461190135e-05, 0.0003187277311499132]
Type: aprilgrid
Tags:
Rows: 6
Cols: 6
Size: 0.088 [m]
Spacing 0.026399999999999996 [m]

cam1

Camera model: pinhole
Focal length: [388.18861658777746, 388.9712538925426]
Principal point: [321.2136971603922, 238.95658293993972]
Distortion model: radtan
Distortion coefficients: [-0.002870215094047284, -0.00471202846643744, 5.440321557997829e-05, -9.458602566352592e-05]
Type: aprilgrid
Tags:
Rows: 6
Cols: 6
Size: 0.088 [m]
Spacing 0.026399999999999996 [m]

IMU configuration
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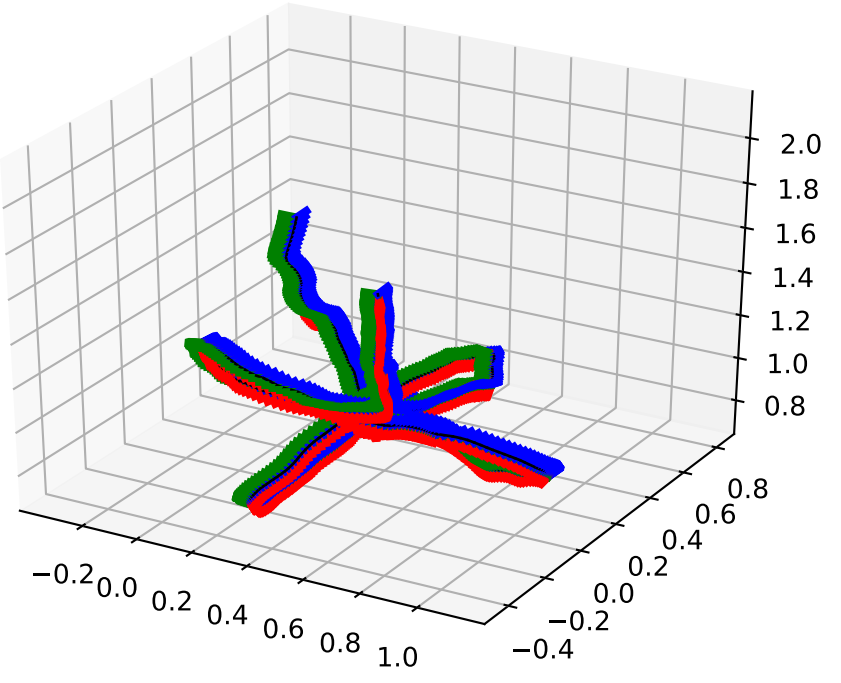
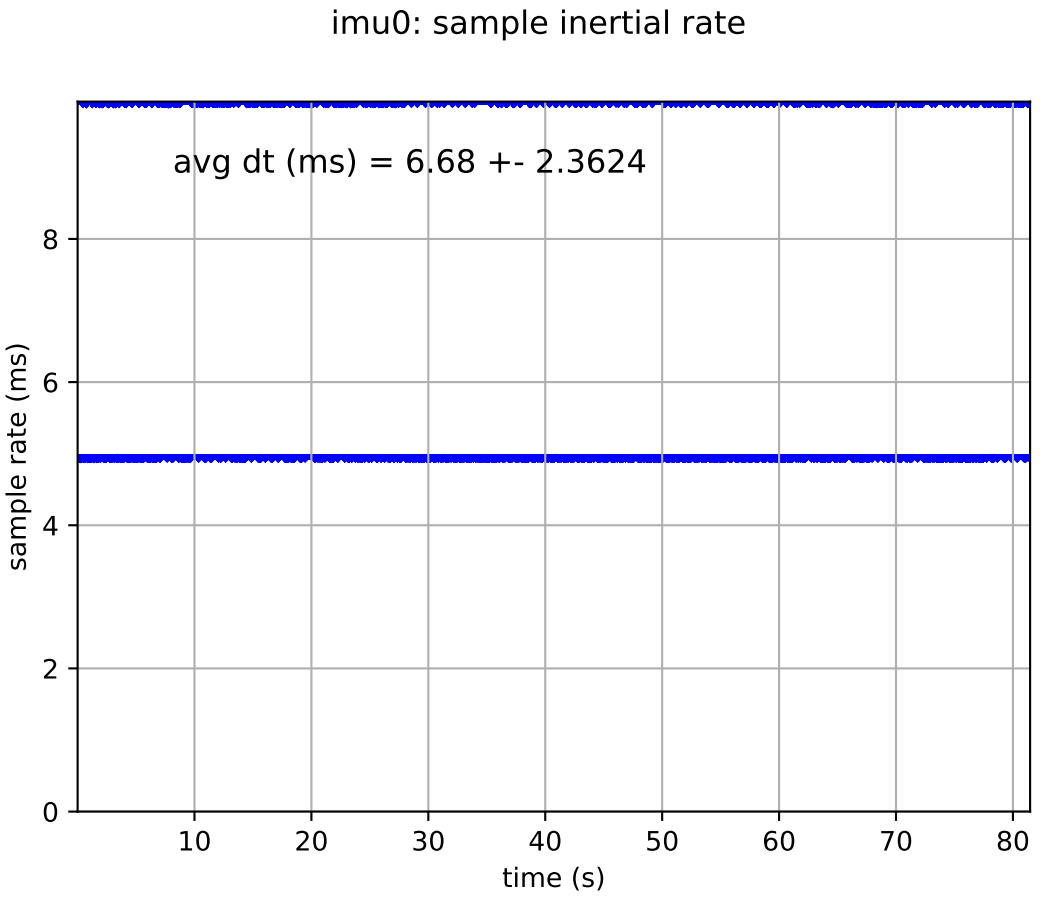
IMU0:

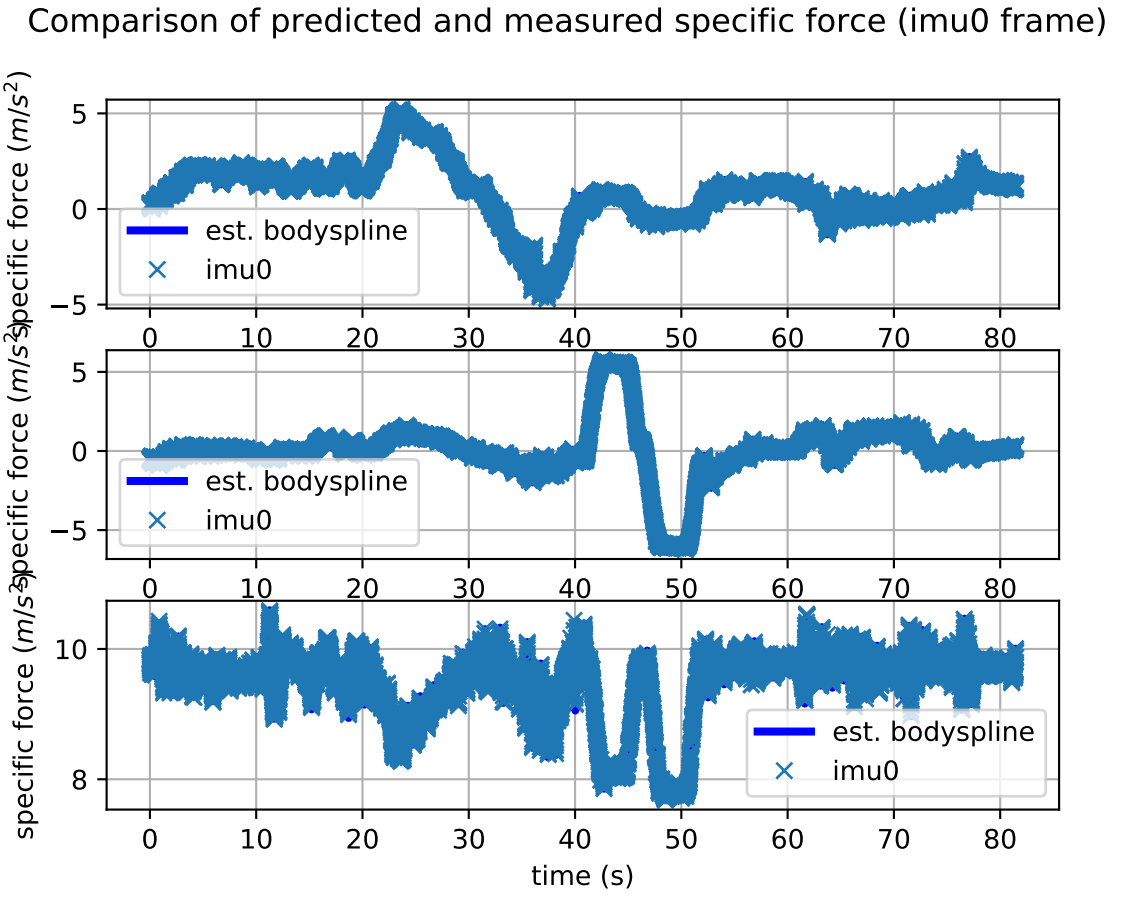
Model: calibrated
Update rate: 149.608877

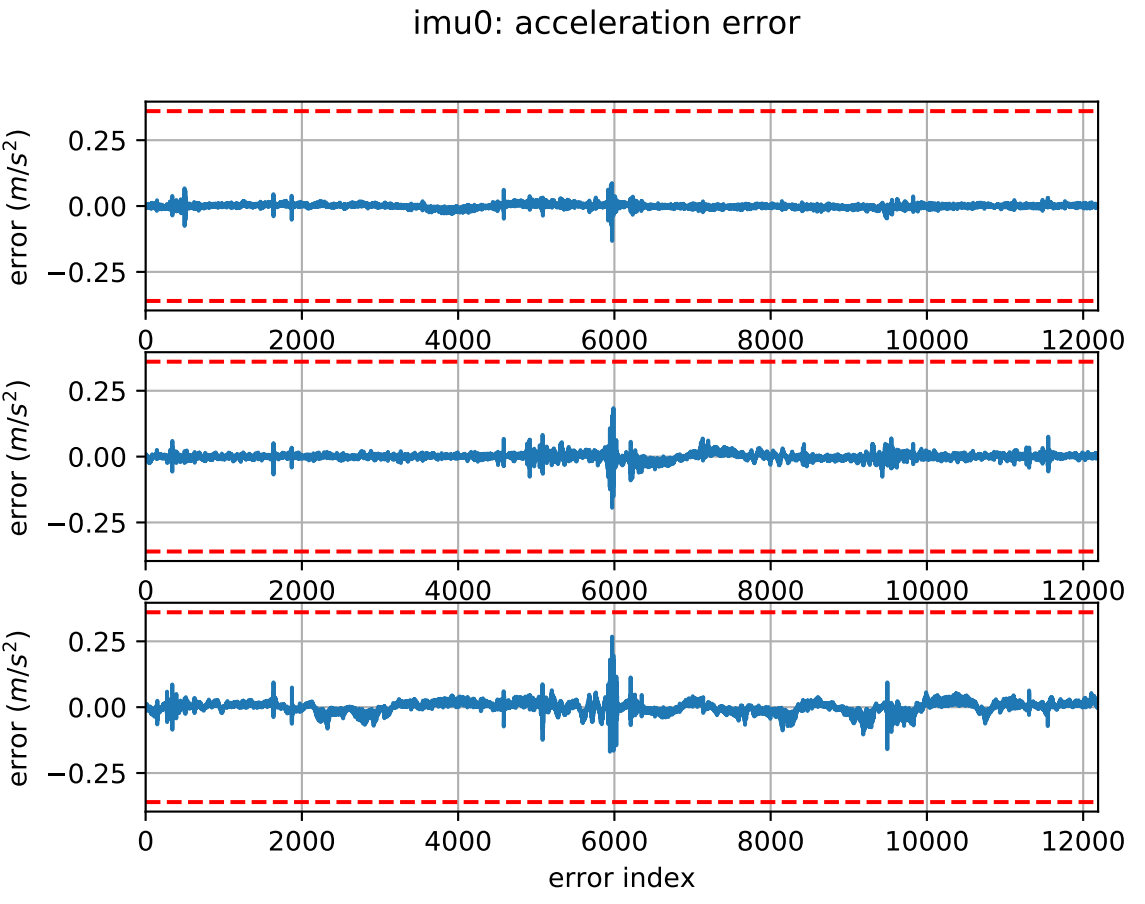
```
Accelerometer:
Noise density: 0.00981989090996197
Noise density (discrete): 0.12011170855056272
Random walk: 0.0005601293637217548
Gyroscope:
Noise density: 0.00043909309937990626
Noise density (discrete): 0.005370754406831473
Random walk: 1.6907369481278766e-05
T_ib (imu0 to imu0)
[[1. 0. 0. 0.]
[0. 1. 0. 0.]
[0. 0. 1. 0.]
[0. 0. 0. 1.]]
time offset with respect to IMU0: 0.0 [s]
```

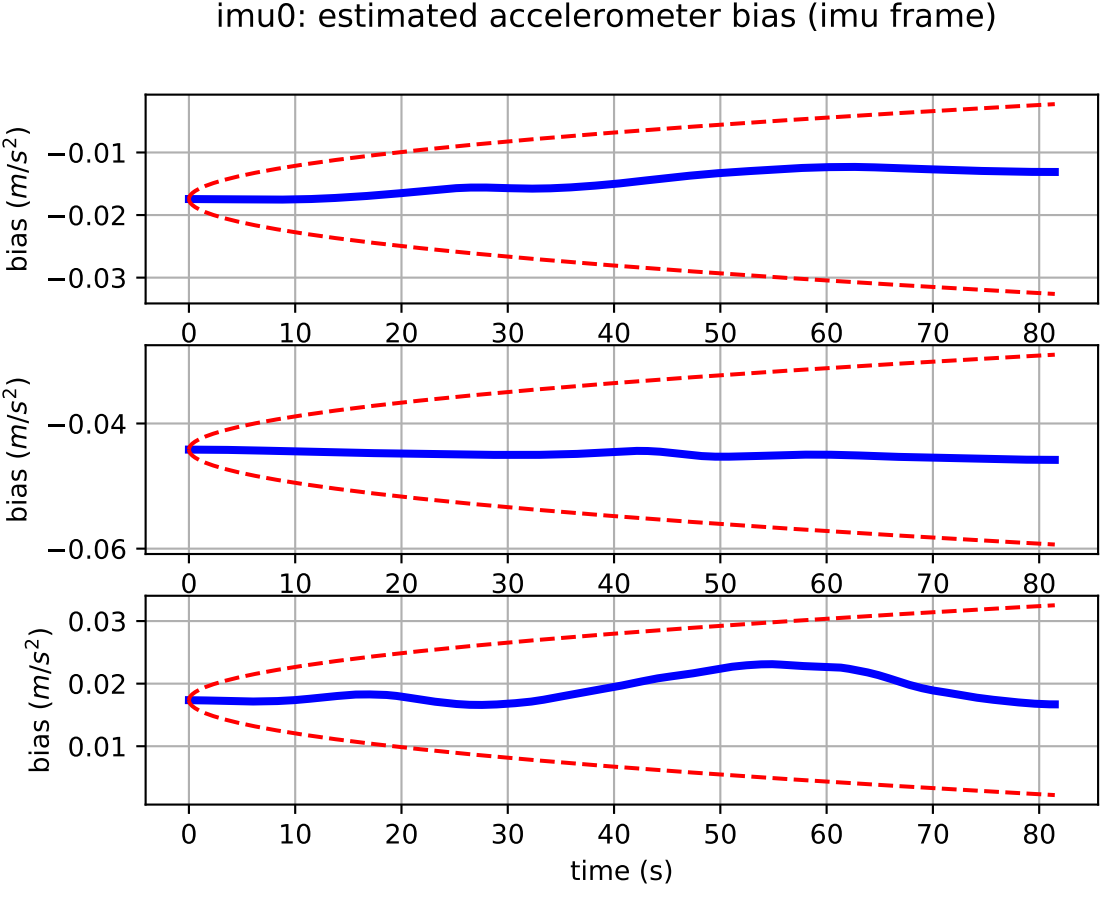
imu0: estimated poses



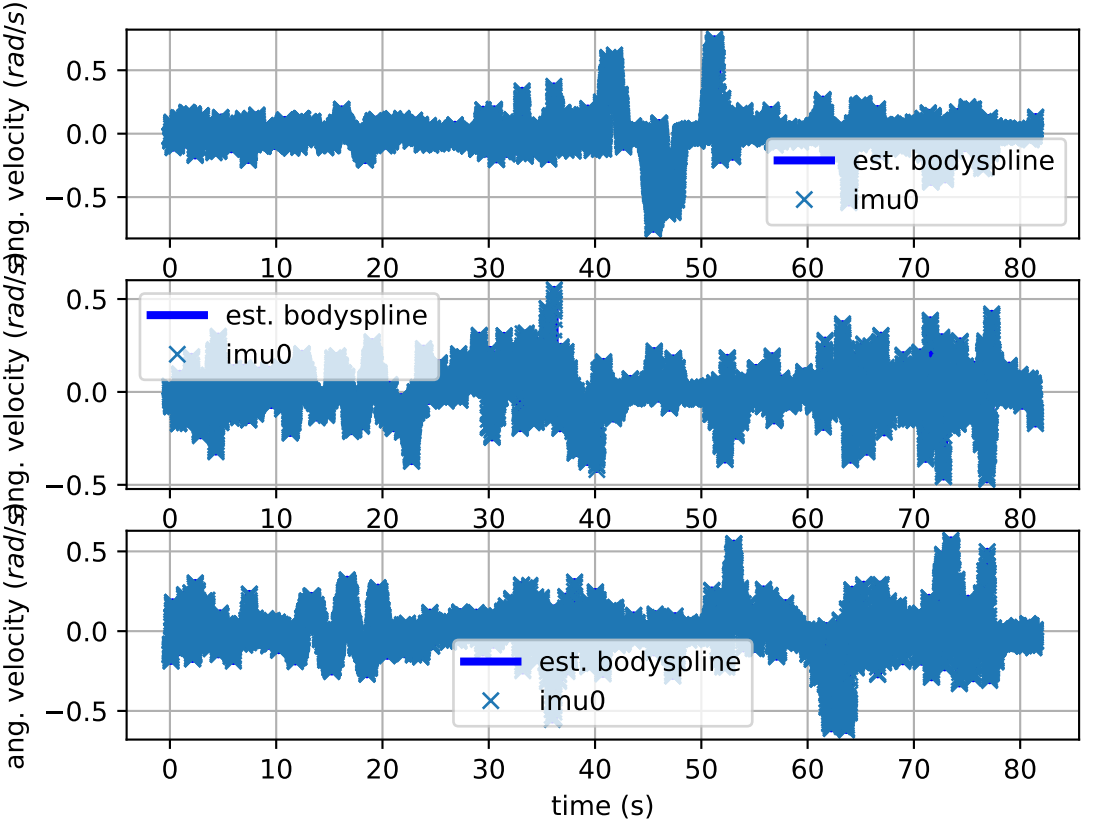

imu0: sample inertial rate
avg dt (ms) = 6.68 +- 2.3624
sample rate (ms)
0
2
4
6
8
10
20
30
40
50
60
70
80
time (s)

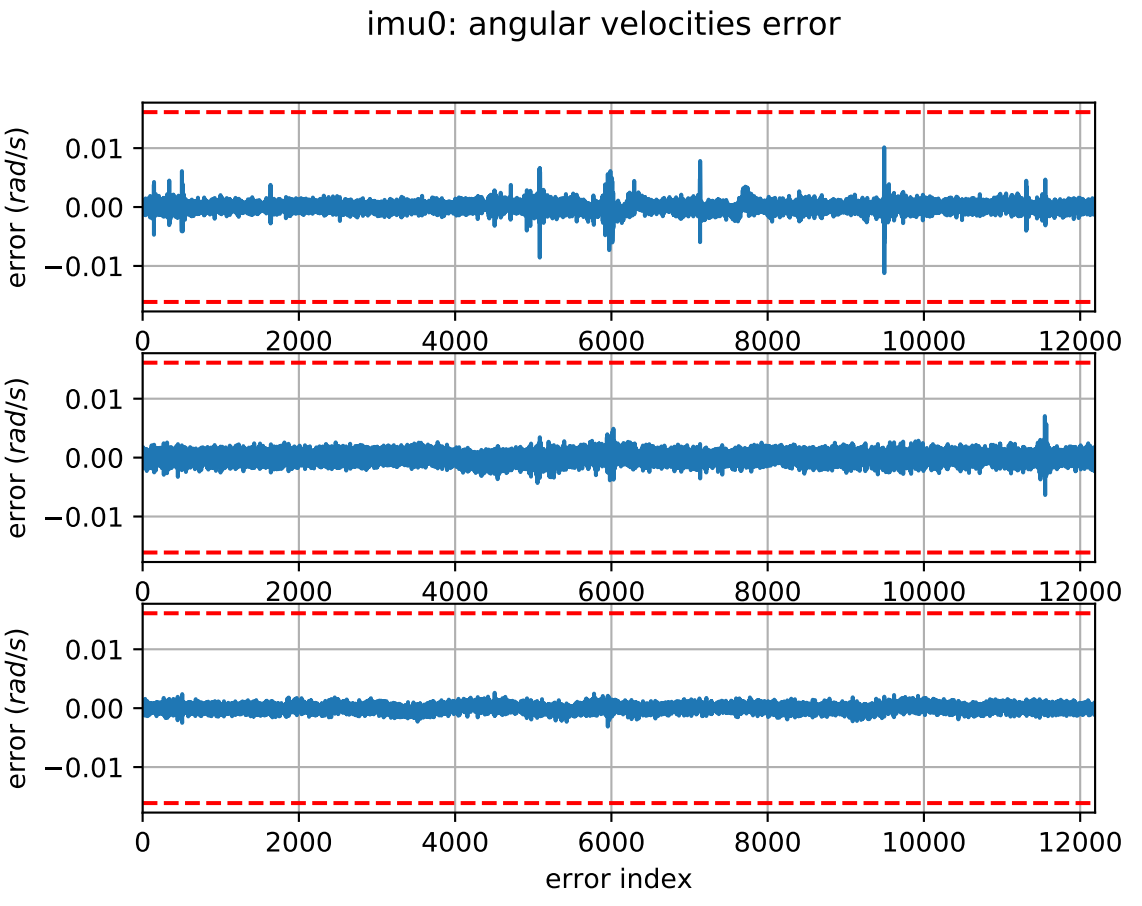

Comparison of predicted and measured specific force (imu0 frame)
specific force (m/s^2)
est. bodyspline
imu0
time (s)

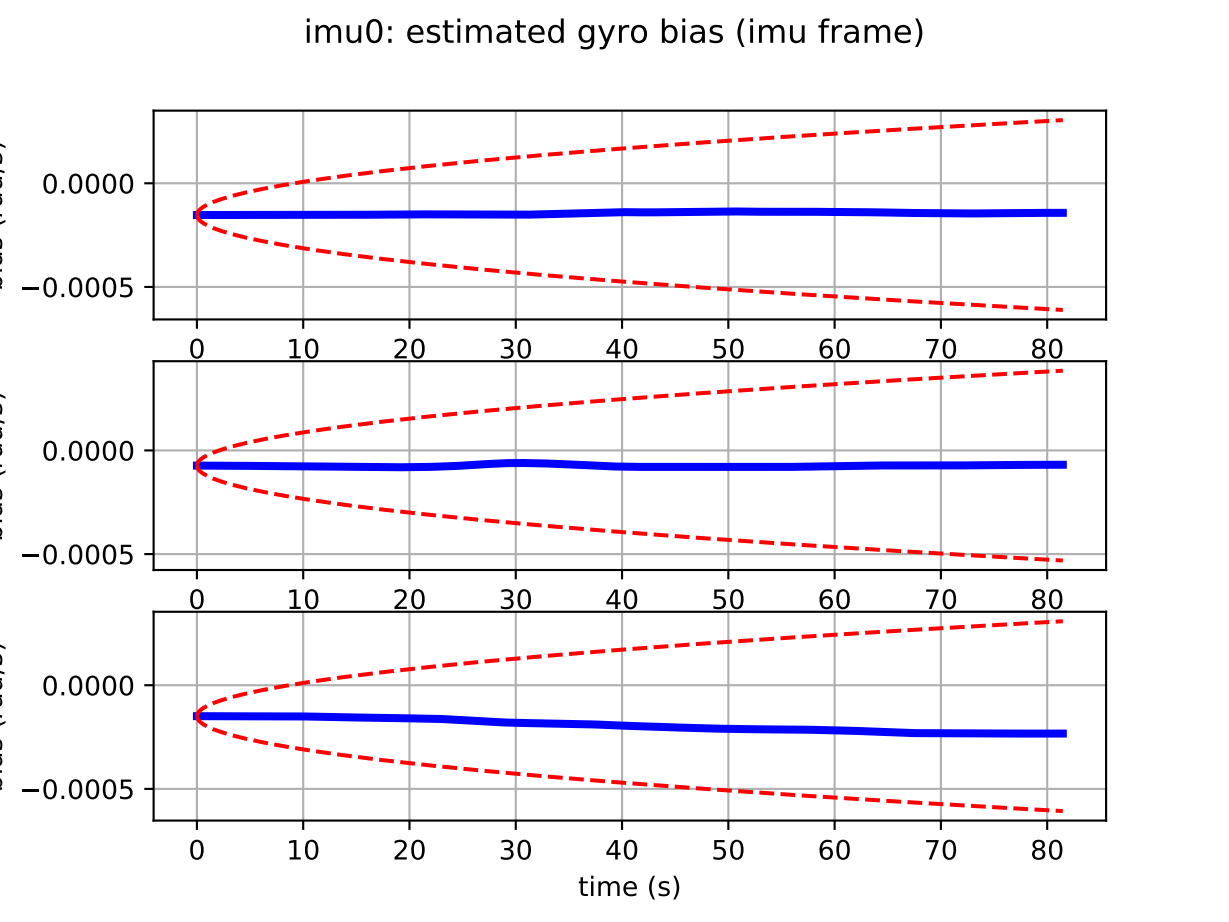

imu0: acceleration error
error (m/s^2)
0.25
0.00
−0.25
0
2000
4000
6000
8000
10000
12000
error index

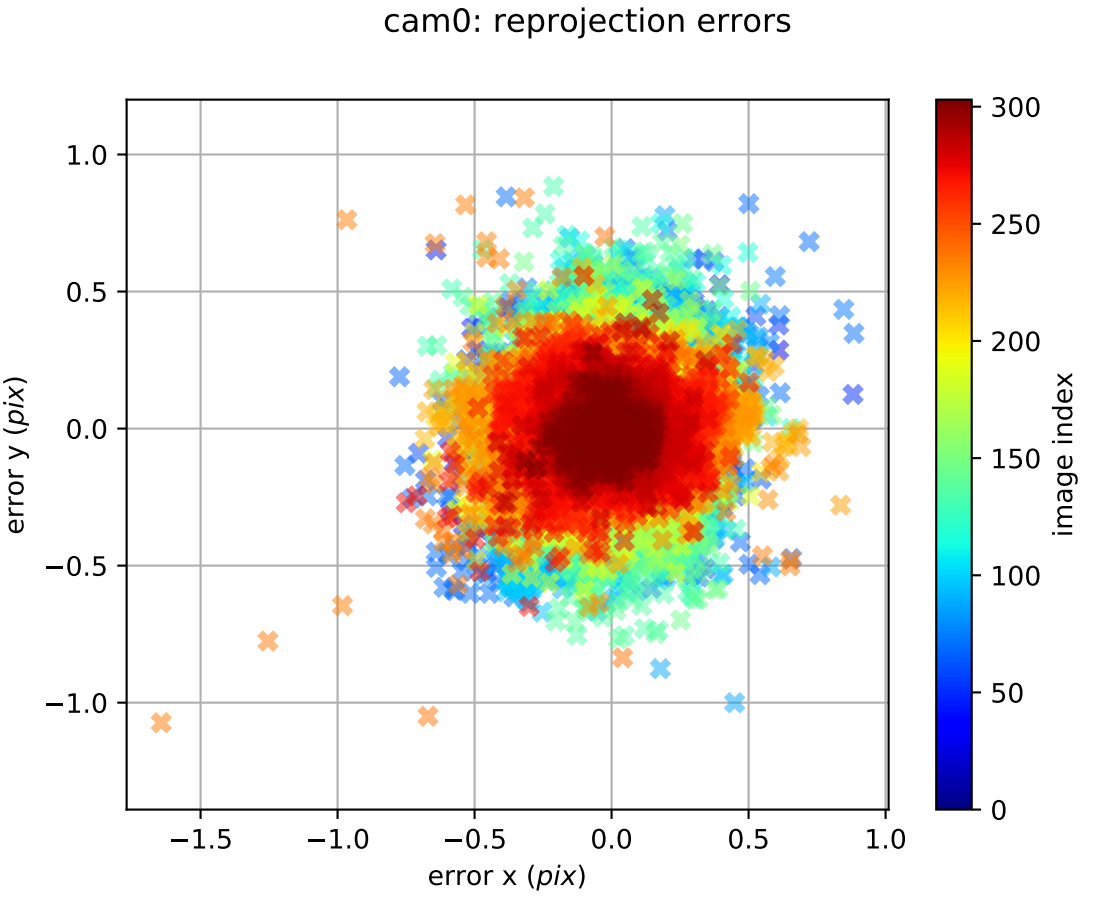

imu0: estimated accelerometer bias (imu frame)
bias (m/s^2)
−0.01
−0.02
−0.03
0
10
20
30
40
50
60
70
80
bias (m/s^2)
−0.04
−0.06
0
10
20
30
40
50
60
70
80
bias (m/s^2)
0.03
0.02
0.01
0
10
20
30
40
50
60
70
80
time (s)

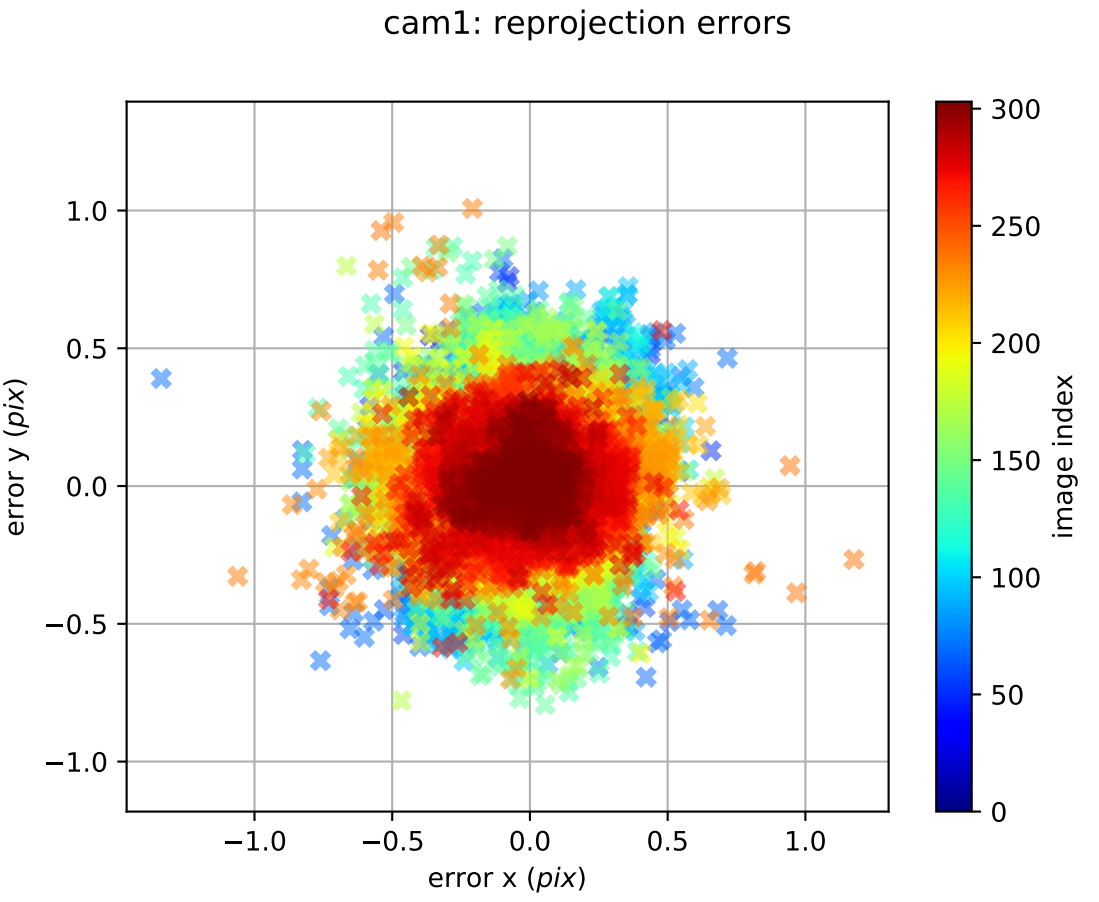
Comparison of predicted and measured angular velocities (body frame)




imu0: angular velocities error
error (rad/s)
0.01
0.00
−0.01
0
2000
4000
6000
8000
10000
12000
error index


imu0: estimated gyro bias (imu frame)
0.0000
−0.0005
0
10
20
30
40
50
60
70
80
time (s)


cam0: reprojection errors
error y (pix)
error x (pix)
image index
1.0
0.5
0.0
−0.5
−1.0
−1.5
−1.0
−0.5
0.0
0.5
1.0
300
250
200
150
100
50
0


cam1: reprojection errors
error y (pix)
error x (pix)
image index
1.0
0.5
0.0
−0.5
−1.0
−1.0
−0.5
0.0
0.5
1.0
300
250
200
150
100
50
0